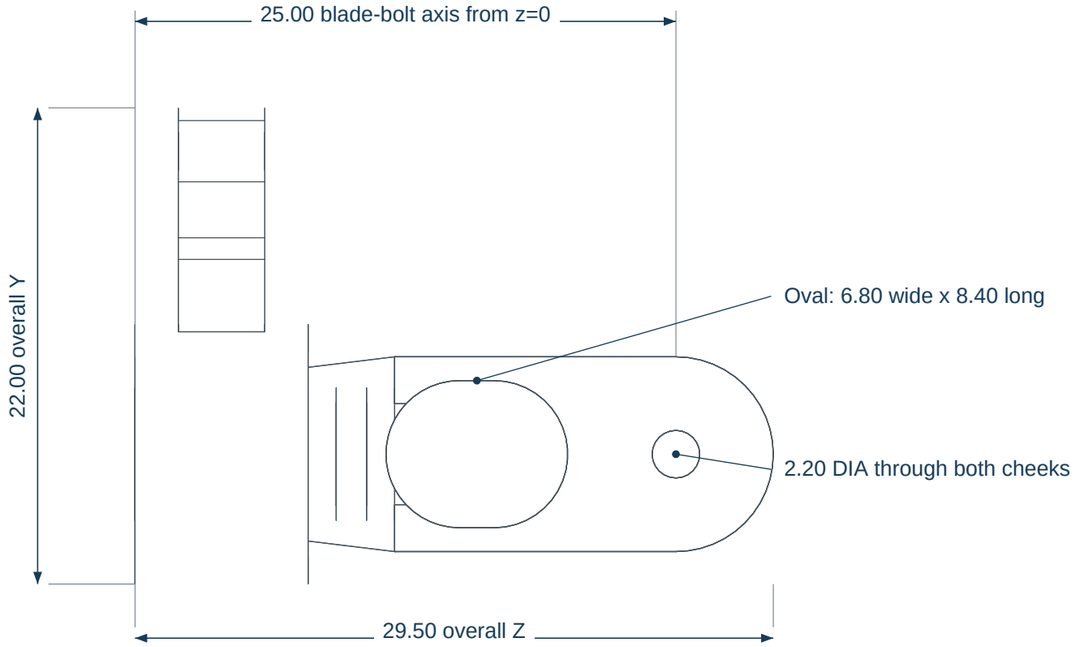
HELICAD / R7

Actual CAD wireframe projections with nominal dimensions

REFERENCE ONLY - NOT FOR MANUFACTURE

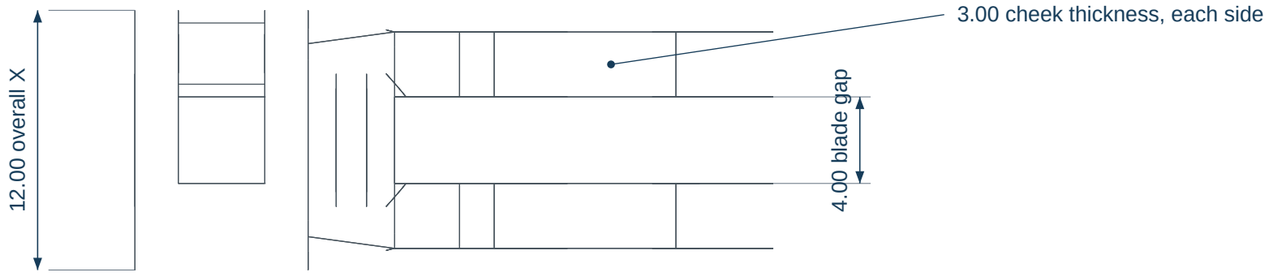
Projections include visible and hidden CAD edges without hidden-line classification. STEP remains authoritative.

VIEW ALONG X / Z horizontal, Y vertical



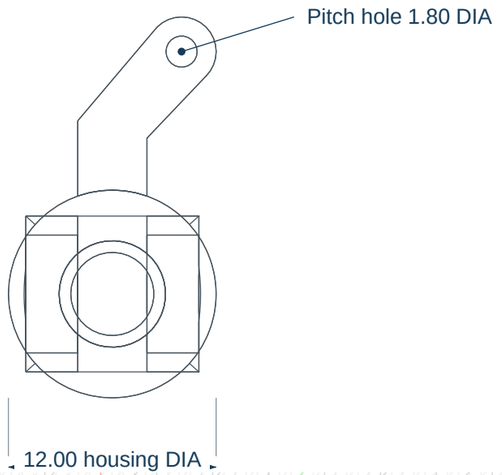
Cheek end R4.50; cheek width 9.00; oval ends z=11.60 and 20.00.

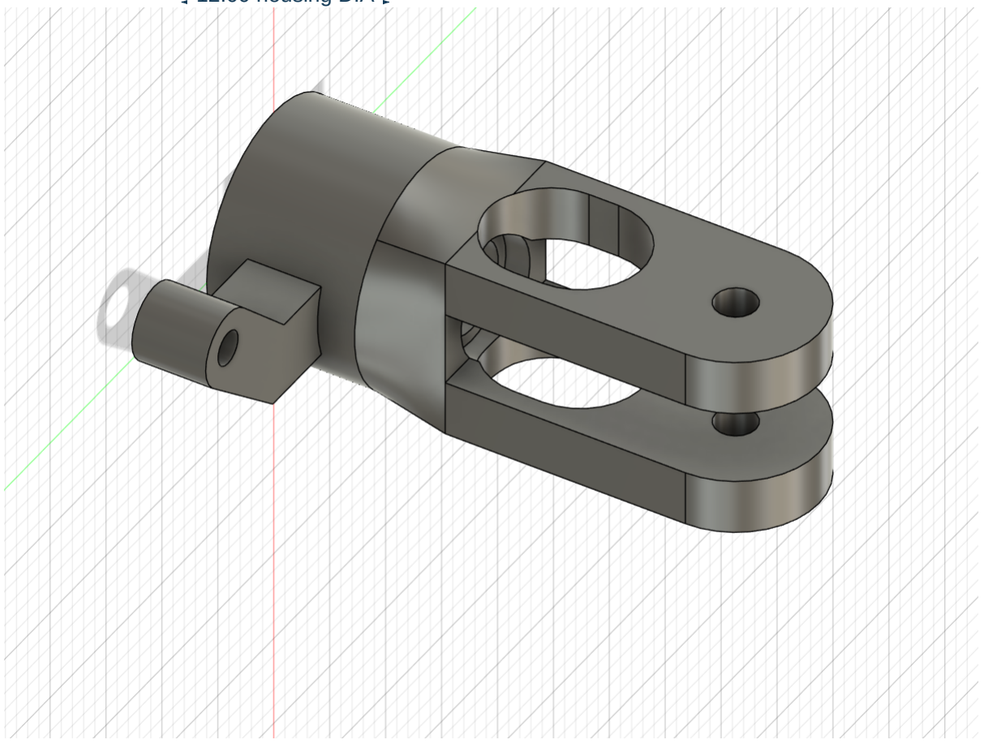
VIEW ALONG Y / Z horizontal, X vertical



Top/blade end: z=29.50. Bottom/bearing opening: z=0.

END VIEW / looking along Z





Isometric view: not to scale.

HELICAD / R7

Dimension schedule, axial stack and tolerance status

REFERENCE ONLY - NOT FOR MANUFACTURE

FEATURE	NOMINAL CAD DIMENSION (mm)	STATUS
Overall envelope	X=12.00; Y=22.00; Z=29.50	Measured CAD bounds
Round outer housing	DIA 12.00; z=0 to 8.00	Nominal / provisional
Housing-to-fork loft	z=8.00 to 12.00	Exact surface in STEP
Loft end envelope	X=10.00; Y=9.00 at z=12.00	Nominal / provisional
Fork cheeks	3.00 thick each; 4.00 clear gap	Nominal / provisional
Fork cheeks X locations	-5.00 to -2.00; +2.00 to +5.00	Nominal / provisional
Fork end / bolt axis	R4.50; axis at y=0, z=25.00	Nominal / provisional
Blade bolt holes	DIA 2.20 through X, both cheeks	Sleeve fit unverified
Oval access	6.80 wide in Y; 8.40 long in Z	Nominal
Oval access location	z=11.60 to 20.00; center y=0	Through both cheeks / X
Bottom bearing pocket	DIA 6.15; z=0 to 9.30	Print allowance only
Terminal shoulder	z=9.30 to 10.70; thickness 1.40	Retention unverified
Shoulder center bore	DIA 4.80	Washer passage restricted
Upper relief cut	DIA 6.15; z=10.70 to 17.60	Interrupted by access slot
Pitch arm thickness	4.00; z=2.00 to 6.00	Nominal / provisional
Pitch hole	DIA 1.80; axis at x=4.00, y=14.00	Parallel to Z; no thread
Pitch arm nose	R2.00; center x=4.00, y=14.00	Exact outline in STEP

TOLERANCES: Not specified. No general +/- tolerance, bearing fit class, coaxiality, finish or mould shrinkage is released.

DIA 6.15 includes an uncalibrated 0.15 diametral allowance over DIA 6 bearings. Do not transfer this fit to aluminium.

All dimensions describe the existing prototype; they are not validated OEM interface requirements.

Material: intended PETG was discussed; CAD last inspected as default Steel. Material/process selection is unresolved.

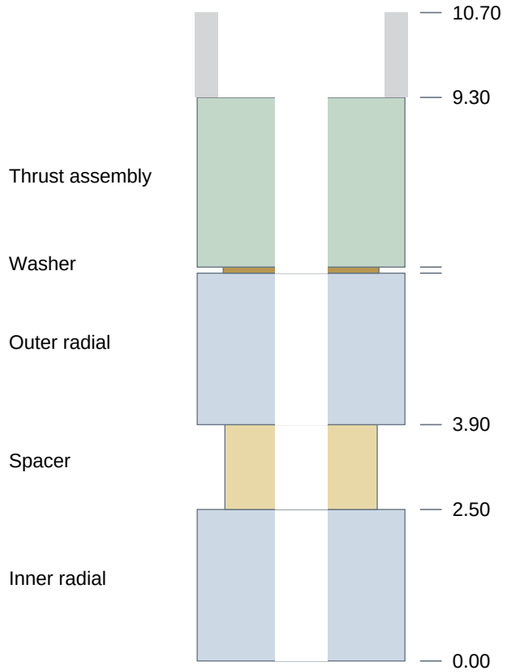
No structural FEA, fatigue test or flight validation. Shaft retention, preload and thrust-race contact require redesign/review.

For complete pitch-arm/loft geometry use the STEP. This PDF is a dimension reference, not a fully toleranced production drawing.

AXIAL STACK / schematic

z=0 at bottom opening; not to scale

Shoulder: 9.30 to 10.70



Washer: z=6.40 to 6.50

Insert from bottom, deepest first:

Thrust > washer > outer radial

> spacer > inner radial